ECE 448 - Lesson 36
HD Radio

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Nominal Schedule

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Lesson	Topics	Notes
22	RF Reverse Engineering	**Final Project Intro**
23	RF Reverse Engineering	
24	RF Reverse Engineering	
25	DTMF	**Remote HW due**
26	DTMF	**Project Proposal due**
27	DTMF	
28	flex day	
29	ADS-B	
30	Project Work Day	
31	ADS-B	**Background Research due**
32	ADS-B	
33	AIS	
34	AIS	
35	Project Work Day	
36	HD Radio	**Design Description due**
37	HD Radio	
38	QAM	**Prototype Description due**
39	Project Day	**GR2 due**
40	Final Project Presentations	**Final Project due**

Lesson 36 Learning Objectives

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36.1 Identify the differences between standard broadcast FM radio and HD radio spectra.

36.2 Identify the modulation and encoding schemes used in various parts of each spectrum.

36.3 Use existing tools to receive and decode HD Radio signals.

HD Radio

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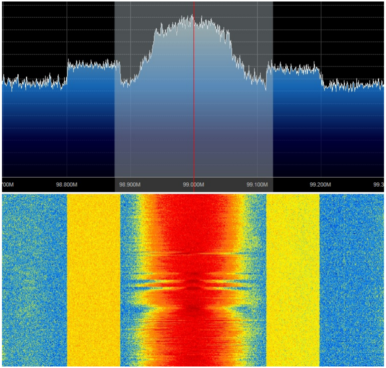
- Allows for transmission of digital & analog signal
- Typical FM analog signal is broadcast 0-100kHz (baseband)
- Digital channels are broadcast as upper/lower sidebands
- Up to 7 digital channels (HD-1, etc.)
- Hybrid: contains analog FM signal (130kHz), digital sidebands
- Hybrid extended: limits analog signal to 100kHz, adds extended sidebands
- All digital



Source: `https://en.m.wikipedia.org/wiki/HD_Radio#/media/File%3AHD_Radio_Romania.jpg`

HD Radio Spectrum

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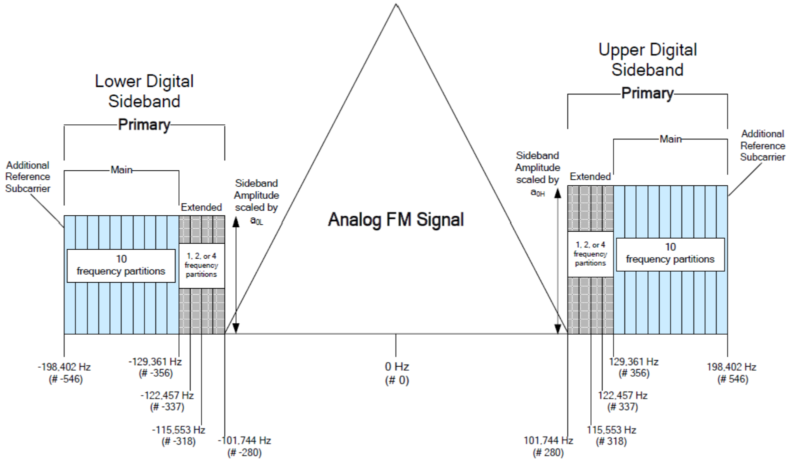

Source: National Radio Systems Committee, 1011s, rev G

Broadcast FM Radio: Spectrum

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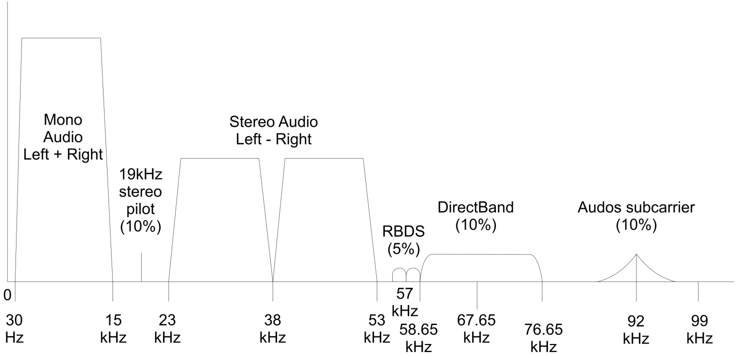

Source: Arthur Murray - `http://en.wikipedia.org/wiki/File:RDS_vs_DirectBand_FM-spectrum2.png`, Public Domain

Broadcast FM Radio: RBDS

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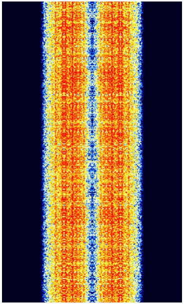
- Known as Radio Data System (RDS) or Radio Broadcast Data System (RBDS)
- BPSK modulation, differentially encoded
- Centered on 57kHz subcarrier
- Each sideband has bandwidth of 2.2kHz
- Data rate is 1187.5 bps, $t_{\text{bit}} = 842\mu s$
- Data contents:
  - Time
  - Text data
  - Programming type
  - Traffic
  - Alt. frequencies
- Tutorial
- gr-rds



Source: `https://www.sigidwiki.com/wiki/File:RBDS.png`

HD Radio: Digital Sidebands

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- Sidebands use Orthogonal Frequency Division Multiplexing (OFDM)
- Power is significantly lower than in center of band
- Each frequency partition consists of 18 OFDM subcarriers (payload data) and a reference subcarrier
- Subcarrier spacing is $\Delta f = 363.4\text{Hz}$.
- Subcarriers are numbered: for Hybrid mode uppger bands, you'll get 356, 375, 394,...,546

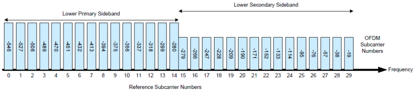


Figure 5-3: Lower Sideband Reference Subcarrier Spectral Mapping

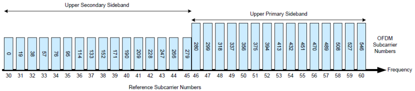
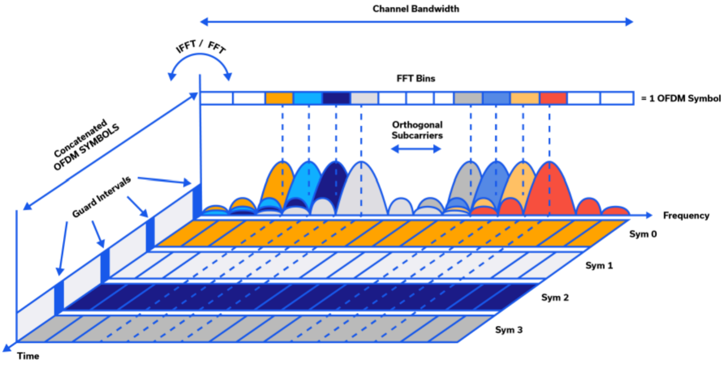


Figure 5-4: Upper Sideband Reference Subcarrier Spectral Mapping

OFDM: Concept



Source:
`https://blog.minicircuits.com/the-basics-of-orthogonal-frequency-division-multiplexing-ofdm/`

OFDM: Concept

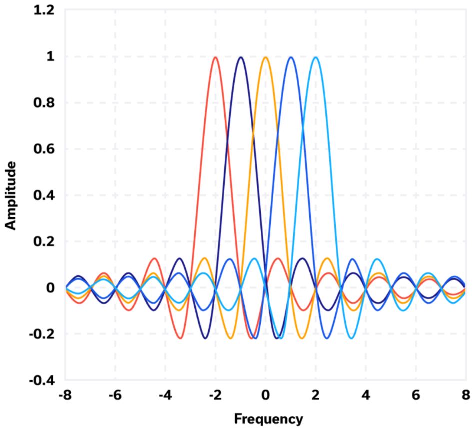


Image source:
https://blog.minicircuits.com/the-basics-of-orthogonal-frequency-division-multiplexing-ofdm/

OFDM: Implementation

- Each subcarrier can be thought of as a "bit" in the OFDM symbol
- Each subcarrier contains information represented by IQ data
- This means each subcarrier can represent a point defined by BPSK, QPSK, or QAM constellations
- Symbols are transmitted over time with "dead time" guard intervals

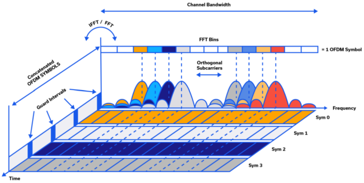
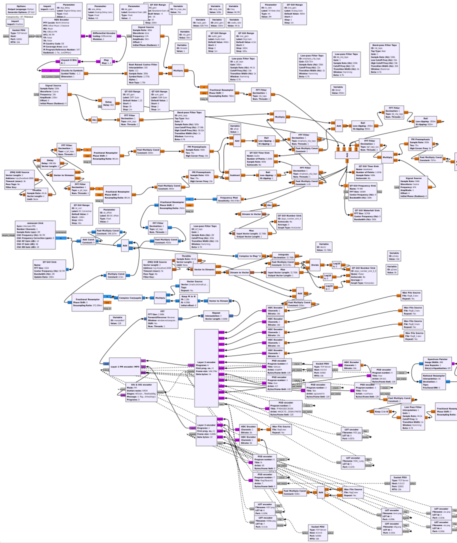


Image Source:
`https://blog.minicircuits.com/the-basics-of-orthogonal-frequency-division-multiplexing-ofdm/`

Tools

- gr-rds
- gr-nrsc5
- NRSC5-DUI



Activities

- Capture HD Radio signal
- Check out the 2023 GRCON CTF challenge (Not Really Superior Clatter-V)
- Parse and decode RDS data (Pysdr.org)
- Recreate OFDM signal

References

- NRSC Site
- Keysight OFDM overview
- SigIDwiki on HD Radio
- PySDR RDS tutorial
- Mini Circuits OFDM overview